META AD PERFORMANCE ANALYSIS

PROJECT OVERVIEW

The **Meta Ad Performance Dashboard** is a critical Business Intelligence (BI) solution designed to deliver timely, actionable insights into the effectiveness and efficiency of all advertising campaigns run across Meta platforms. This project provides a unified view of performance metrics, empowering marketing and campaign managers to make data-driven decisions.

This **Meta Ad Performance Dashboard** that tracks the effectiveness of ad campaigns across key KPI's such as ***impressions, clicks, engagements, conversions, and budget***. It provides a complete funnel view-from awareness to engagement to purchases-along with demographic, geographic, and time-based insights.

DATA ARCHITECTURE AND PROCESSING

About the Data

This dataset represents Meta Ads Performance Data, covering campaigns, ads, user demographics, and ad interaction events. It is modelled after how Facebook/Instagram (Meta) ad platforms capture data.

The purpose of this dataset is to analyse advertising performance, optimize targeting, and measure ROI (Return on Investment) through KPIs such as:

- Impressions (how often ads are seen)
- Clicks (engagement with ads)
- Purchases (conversions)
- CPM, CPC, CTR, and ROAS (efficiency metrics)
- Audience insights (demographics, location, interests)

This dataset is ideal for building a Power BI Dashboard to evaluate campaign effectiveness, budget utilization, and user engagement patterns.

Data Processing (Power Query / M-Code)

The source data, having been downloaded in a pre-cleaned state from a public repository, required minimal data integrity work. Transformation efforts were focused exclusively on optimizing the data for the Power BI model.

Key Transformation Steps:

1. **Source Data Load:** All four CSV files were loaded using the **Import** mode.
2. **Key Standardization:** Foreign key columns (ad_id, campaign_id, user_id) were explicitly set to the correct data types (**Integer** or **Text**) to guarantee relationship integrity.
3. **Time Series Preparation:** The timestamp column in ad_events.csv was ensured to be a proper **DateTime** type.
4. **Date Table Generation (DAX):** The central **Calendar Table** was generated using **DAX** to create granular time fields (including **Week** and **Hour** dimensions) essential for advanced time-based analysis.

Data Modeling (Star Schema)

The model is structured as a **Star Schema**, centered around the ad_events Fact Table. This architecture is best practice for BI dashboards, ensuring optimal query performance and measure simplicity.

Model Diagram:

Key Relationships:

All relationships are **One-to-Many** (* to 1) with a Single cross-filter direction, flowing from the Dimension tables to the central Fact table.

Fact Table	Dimension Table	Relationship Field	Cardinality
ad_events	ads	ad_id	$\ast : 1$
ad_events	users	user_id	$\ast : 1$
ad_events	Calendar	timestamp/Date field	$\ast : 1$

Fact Table	Dimension Table	Relationship Field	Cardinality
ads	campaigns	campaign_id	$\ast : 1$

Data Set Description

This section details the schema for each dataset, outlining the columns used for analysis.

ad_events.csv (Fact Table)

This is the core transactional table, recording user-level actions.

Column Name	Data Type	Description
event_id	Integer	Unique identifier for the recorded event.
ad_id	Integer	**Foreign Key (FK)** to the ads table.
user_id	Text	**Foreign Key (FK)** to the users table.
timestamp	DateTime	Exact time and date of the user interaction. **Used to create Date Dimension.**
event_type	Text	The nature of the interaction: **Impression**, **Click**, **Like**, **Share**, **Purchase** (Conversion).

ads.csv (Dimension Table)

This table holds static information about the ad creatives themselves.

Column Name	Data Type	Description
ad_id	Integer	**Primary Key (PK)**. Links to ad_events.
campaign_id	Integer	**Foreign Key (FK)** to the campaigns table.
ad_platform	Text	The channel where the ad ran (Facebook, Instagram).
ad_type	Text	The creative format (Video, Stories, Carousel, Image).
target_gender	Text	The gender audience targeted by the ad.
target_age_group	Text	The age range targeted by the ad.
target_interests	Text	Interests targeted by the ad.

campaigns.csv (Dimension Table)

This table defines the high-level business context and budget for each campaign.

Column Name	Data Type	Description
campaign_id	Integer	**Primary Key (PK)**. Links to ads.

Column Name	Data Type	Description
name	Text	Descriptive campaign name (e.g., "Campaign_1_Launch").
start_date	Date	Official start date of the campaign.
end_date	Date	Official end date of the campaign.
total_budget	Float	The total amount of money allocated for the campaign.

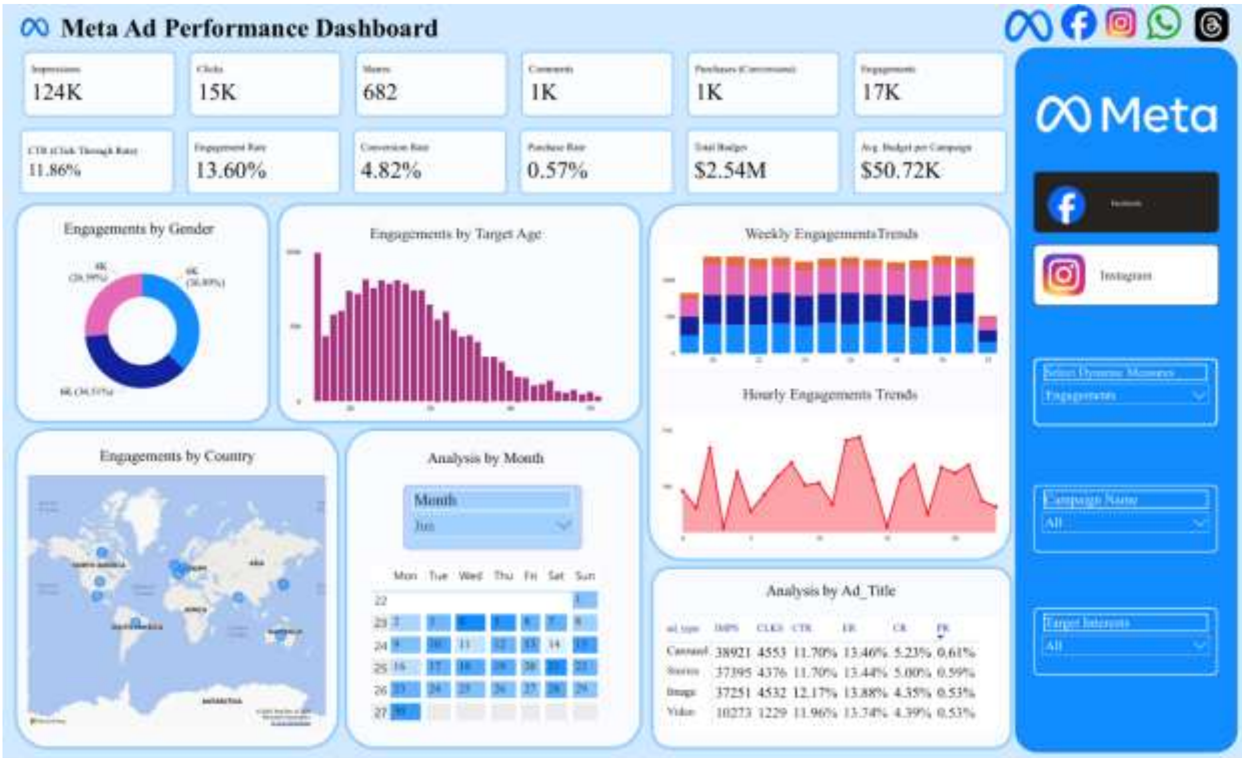
users.csv (Dimension Table)

This table contains demographic information about the users.

Column Name	Data Type	Description
user_id	Text	**Primary Key (PK)**. Links to ad_events.
user_gender	Text	The gender reported by the user.
age_group	Text	User's age classification (e.g., 25-34, 35-44).
country	Text	User's country of residence.
location	Text	User's specific location/city.
interests	Text	Comma-separated list of the user's interests.

Report / Dashboard Design

The dashboard is structured into logical pages, utilizing the unique time granularity to provide deeper insights into campaign operations.



Walk through the Key Measures

At the top of the funnel, the ads generated ***216K impressions*** and ***25.4K clicks***, giving a very high ***CTR of 11.76%***. This is well above the industry average, which tells me the ad creatives and targeting were very effective in attracting attention.

The engagement rate is also strong at ***13.56%***, showing users are interacting with the content. However, when we move down the funnel, only ***1.3K purchases*** were made, giving a ***conversion rate of 5.21% from clicks and a purchase rate of 0.61% from impressions***. This indicates a big drop in efficiency from engagement to purchase.

Key Takeaways:

The ads are good to generating awareness and engagement, but the purchase funnel is leaking heavily – likely due to landing page experience, audience mismatch, or weak offers.

Audience Analysis

Looking at demographics, females (43%) engage more than males (22%), and the 18–30 age group drives the majority of interactions. This shows the core audience is young females.

From a geographic perspective, India and Brazil are the biggest engagement markets, while Germany and the UK likely represent higher-value audiences with stronger purchasing power. So, campaigns should be tailored differently for high-volume vs. high value regions.

Time & Seasonality

The dashboard shows consistent weekly engagement, but hourly trends peak in the afternoon and evening hours. This suggests that ads should be scheduled and budget weighted towards those times to maximize ROI.

The calendar highlights certain days (19th–21st, 25th–27th) with spikes in engagement, which could be linked to promotions or campaign launches. This indicates that event-based campaigns drive higher performance.

Recommendations

In summary:

1. Strong awareness & engagement, but low purchase efficiency → optimize landing pages, retargeting, and offers.

2. Target audience = young females, 18–30, in India & Brazil → refine campaigns accordingly.

3. Best formats = Video and Stories → increase spend here.

4. Best times = afternoons & evenings → schedule ads accordingly.

5. Geography → volume from India/Brazil, value from Germany/UK → segment strategies.